

IOSG Best 15 ETF

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Introduction

The IOSG Best 15 ETF is a long-only, zero-leverage weighted index product targeting the top 15 most valuable secondary crypto assets. The construction encodes IOSG Ventures’ decade of crypto investing experience into a transparent, rules-based methodology for capturing risk-adjusted alpha while balancing volatility.

Although the broader crypto market has corrected meaningfully since the 2021–2022 cycle peak, a number of high-quality tokens have emerged as durable secondary-market exposures as the industry matures into a post-bubble phase. The ETF product offers institutional and retail investors a transparent, monthly-rebalanced vehicle with disciplined drawdown management and a clear methodology for asset selection.

By combining an objective, market-cap-ranked universe with tradability filters (risk flags) and an adjusted minimum-variance portfolio construction, the ETF aims to deliver diversified exposure with performance comparable to — or exceeding — Bitcoin, while constraining concentration and protecting against tail liquidity events.

Universe Selection

Preliminary Requirement

To enter the index universe, a token cannot be:

1. A stablecoin, wrapped, staked, pegged, bridged, or tokenized asset
2. A memecoin or a gaming token

This filter is applied monthly at the rebalance date against the top 400 tokens by market capitalization. The resulting pool is the **Start Universe**, which is then narrowed further via the risk flags below.

Risk Flag

On top of preliminary filtering, three risk flags are applied to avoid token pitfalls that historically underperform.

1. Liquidity

A two-step score based on each token’s **30-day median trading volume**, benchmarked against the volume distribution of the **Top 100 tokens by market capitalization** (the reference group used in both steps).

Step 1 — Boolean tradability flag. Within the Top 100 by market capitalization, compute each token’s 30-day median trading volume and identify the 25th percentile of that volume distribution. Any token whose volume falls below this p25 threshold is marked as untradable (Liquidity Flag = False).

Step 2 — Tradable tier scoring. Tokens that pass Step 1 are scored on a 2–5 scale using volume percentiles **recalculated within the tradable group only**:

Score	Condition
5	Volume > 75th percentile (within tradable)
4	Volume > 50th percentile (within tradable)
3	Volume > 25th percentile (within tradable)
2	Volume $\leq$ 25th percentile (within tradable)
1	Liquidity Flag = False (volume below the p25 of the Top 100 by market capitalization)

2. Inflation Expectations

Score reflects projected circulating supply increase over the **next 12 months**, sourced from DefiLlama emissions data.

Score	Condition
5	$\leq$ 5% projected supply increase (minimal dilution)
4	> 5%, below 25th percentile within the high-inflation subset
3	> 5%, 25th–50th percentile
2	> 5%, 50th–75th percentile
1	> 5%, 75th–90th percentile
0	> 5%, above 90th percentile (extreme dilution risk), or no DefiLlama emissions data available

The 5% threshold reflects a common annual inflation benchmark below which dilution impact on holders is considered minimal. Tokens lacking emissions data are penalized to score 0 by default — unknown vesting risk is treated as the worst-case outcome rather than silently assumed to be zero.

3. Token Value Accrual

Identifies whether an explicit on-chain mechanism transfers value to token holders, or whether the token is purely governance-bearing.

Score	Definition
5	Active programmatic value flows on-chain (buybacks, burns, dividend distributions)
4	Clear value accrual framework designed but no active distribution yet; no competing equity, fees, or off-token capture; token explicitly defined as the sole vehicle for value accrual
3	Partial value flows with competing value sinks; misalignment between lab entities and DAO interests; on-chain burn does not capture main revenue source
2	Partial flows with competing value sinks plus opaque/discretionary mechanism (off-chain or non-programmatic dividend distributions)
1	Complete value extraction before token holders; all fees go to labs with zero to DAO; no meaningful accrual beyond governance

Boolean Exchange Listing Flags

Three additional Boolean flags are computed at the rebalance date:

- Listed on Binance $\in \{\text{True}, \text{False}\}$
- Listed on Coinbase $\in \{\text{True}, \text{False}\}$
- Listed on Binance OR Coinbase $\in \{\text{True}, \text{False}\}$

Tokens that satisfy the preliminary requirements and clear the Risk Flag thresholds form the eligible universe. The ETF further narrows this universe through the constraint structure described in the Index Construction section.

Index Construction

Overall Design

The ETF is executed as a **long-only, zero-leverage** strategy with monthly rebalancing. The mathematical objective and the per-stage covariance adjustments are described in the *Appendix: Portfolio Optimization*.

The ETF is structured around a single quantitative engine:

- **Risk Factor (Rt)** — selects the **top 15 crypto assets** by previous-day market capitalization (subject to risk flags), then derives weights via adjusted minimum-variance optimization with tiered per-asset caps and group constraints (see *ETF Constraint Structure* in the Appendix).
- **Technical Signals shrink the diagonal variance of high-momentum assets** (top quantile) and **inflate it for low-momentum assets** (bottom quantile). Lower variance $\rightarrow$ MinVar allocates more weight; higher variance $\rightarrow$ less weight. Net effect: the portfolio tilts toward shorter-term momentum without overriding the Rt universe selection or constraint structure.

The embedded assumption is that **assets with sufficient market capitalization and clean risk flags** remain the primary gatekeepers of asset legitimacy, while technical signals fine-tune allocation within that vetted pool.

- **Liquidity Penalty inflates the diagonal variance** of low-volume tokens (bottom 50% by 30-day median volume) by up to 3×. Higher variance → MinVar allocates less weight. Net effect: illiquid names are pushed away from the upper bounds of their tier caps, regardless of their pure variance ranking.

Cash Overlay

The final ETF allocation is:

$$\text{ETF} = 80\% \text{ Rt} + 20\% \text{ Cash}$$

Backtesting demonstrates that **the ETF delivers superior risk-adjusted returns with a cash overlay than without**. On the production configuration (Top 15 MCAP, Binance or Coinbase listed, Jan 2020 – Apr 2026), adding a 20% cash reserve improves every dimension of risk-adjusted performance:

Metric	Effect of 20% Cash Overlay
Sharpe Ratio	+9% higher
Max Drawdown	13% smaller
Annualized Volatility	21% lower

The cash buffer materially improves drawdown management without meaningfully sacrificing upside in directional bull regimes. A **fixed 20%** allocation is chosen as the operating point: it provides defensive positioning during sharp drawdowns while preserving sufficient capital efficiency for upside participation. The allocation is static (not dynamically managed) to keep the ETF rules-based and avoid discretionary timing.

Cash is deployed in institutional-grade tokenized money-market vaults (Superstate / Securitize, US Treasury-backed) — providing additional yield on top of the risk-reduction benefit, with no lock-ups and no counterparty credit risk.

Rt

Rt is the portfolio-optimization methodology designed to construct a 15-asset crypto portfolio that balances risk minimization with liquidity considerations and short-term technical signal awareness.

The framework implements an **adjusted minimum-variance** approach: the liquidity penalty and technical signals are incorporated directly into the covariance matrix diagonal (via variance multipliers), and an entropy cost term is added to the objective function to prevent extreme concentration.

The core innovation addresses a critical limitation of standard minimum-variance optimization: the tendency to over-allocate to low-volatility but illiquid assets, which can lead to execution difficulties and hidden tail risks.

Rt Universe

On top of the Start Universe, additional risk flags ensure only tradable tokens with genuine value enter the optimization:

- Tokens must be listed on Binance or Coinbase
- Exclude Liquidity score = 1
- Exclude Inflation Expectations score = 0
- Exclude Token Value Accrual score = 1

The ETF then selects the **top 15 crypto assets by previous-day market capitalization** as the default universe. Market-cap ranking serves as a proxy for adoption and network effects; liquidity is enforced separately via the Liquidity Penalty (Stage 3 of the Optimization Pipeline in the Appendix) rather than at the universe-selection step.

Crypto-Adjacent Stocks Overlay (Optional Module)

A crypto-adjacent stocks overlay module can be activated to incorporate 3 equity-linked assets (such as Exchanges, Stablecoin Issuers, or Digital Asset Treasury Companies) that share economic exposure to crypto market cycles. When the overlay is active, it replaces the bottom 3 crypto assets, keeping the total at 15 constituents (**top 12 crypto + 3 stocks**). The overlay is **off by default** in the base ETF specification and can be turned on as a separate product variant without modifying the underlying Rt machinery.

Cycle Signal Overlay (Optional Module)

A cycle signal overlay module can be activated to dynamically scale the cash allocation rather than holding the 20% buffer static. The model combines **59 systematic signals** spanning on-chain data, traditional financial markets, and US macro indicators, aggregated each morning into a composite regime score that drives a daily-updated cash level between **0% and 50%**. The pipeline is fully rules-based — no discretionary overrides — and is designed to compress drawdowns during cycle reversals while restoring full crypto exposure during clear bull regimes. The overlay is **off by default** in the base ETF specification and can be activated as a separate product variant without modifying the underlying Rt machinery.

Rt Weighting Methodology

Rt weights are derived by passing the 15-asset universe through the **Optimization Pipeline** (Stages 1–5) described in the *Appendix: Portfolio Optimization*. The pipeline applies, in order: covariance estimation, trace-normalized L2 ridge, liquidity penalty (variance multiplier on the bottom-50% by 30-day median volume), technical signal adjustment (vol-adaptive top-set / bottom-set diagonal multiplier), and entropy calibration. The full SLSQP problem is then solved in a single call subject to the per-asset bounds, group floors, and group caps defined in *ETF Constraint Structure* (Appendix).

Key Considerations

1. **Why adjust minimum variance?** Unadjusted minimum variance heavily weights low-volatility, low-correlation assets, often concentrating in extremely illiquid names. Variance multipliers along the covariance diagonal correct this bias.
2. **Why Binance or Coinbase as the listing filter?** Both are tier-1 venues with deep secondary-market liquidity; combining them broadens the universe without admitting tokens that only trade on niche or low-quality exchanges, which would be impractical for an ETF wrapper under monthly rebalancing turnover.
3. **Why previous-day data?** Ensures only information available at the decision moment is used, avoiding look-ahead bias.

Technical Signals

Technical signals are a complementary adjustment layer on top of R_t — **not a standalone factor**. They influence allocation only through the variance-multiplier mechanism described in *Stage 4 of the Optimization Pipeline* (Appendix); they cannot override the R_t universe selection or constraint structure. Market capitalization ranking and risk flags remain the primary determinant of which assets enter the portfolio and at what magnitude; the liquidity penalty (Stage 3) and technical adjustment (Stage 4) are subordinate weighting refinements within that vetted pool.

Data Sources

CoinGecko market capitalization, token price, and trading volume series.

Technical Factor Universe

The technical layer draws from a pool of **173 indicators** systematically grouped into ten families, each representing a distinct market intuition: Mean Reversion, SMA Cross, EMA Cross, RSI, MACD, Bollinger Bands, Volume Spike, Price Appreciation, Z-Score vs MA50, and Momentum. Each family is described in detail in the *Indicator Family Specifications* section below.

Two-Layer Selection Methodology

Selection is governed by three core principles: **objectivity** (quantitative metrics, not subjective judgment), **robustness** (strict separation of training and testing periods, no look-ahead bias), and **diversification** (signals drawn from ten distinct families).

The selection process operates in **two layers** — one inside each family, one across families — so that the engine separately controls *which variant of an indicator to use* and *which families are robust enough to deploy*.

Layer 1 — Within-family indicator selection

For each family, every rolling 12-month training window evaluates all variant columns and picks the best performer based on:

- **Rank IC** — correlation between factor-value ranks and forward-return ranks (cross-sectional ranking accuracy, robust to outliers)
- **Q5–Q1 Sharpe** — Sharpe ratio of (top-quintile minus bottom-quintile) factor returns (directly reflects monetizable predictive power)

The selected variant remains **fixed for the subsequent 3-month test period** without rotation, preventing forward-looking bias. At each monthly rebalance during that window, this fixed variant is the active parameter for its family.

Layer 2 — Across-family deployment selection

A family being internally well-tuned (Layer 1) does not guarantee it is robust enough to deploy. The family-level decision uses a separate composite score:

- **70% Q5–Q1 Sharpe** — of the family’s rolling backtest using its Layer-1-selected variant (in-sample monetizability)
- **30% Robustness composite** — blend of three components:
 - **Rolling-window Sharpe**
 - **t-statistic**
 - **Deflated Sharpe Ratio (DSR)**
- **Additional DSR penalty** — further correction for the multiple-testing bias arising from evaluating many families simultaneously

This two-layer structure cleanly separates concerns: Layer 1 tunes *which version of an indicator* to use inside each family; Layer 2 decides *which families* survive a multiple-testing-corrected robustness check and enter the technical adjustment at all.

Metrics Reference

Metric	Layer	Definition	Role
Rank IC	L1	Correlation between factor-value ranks and forward-return ranks; range -1 to +1	Picks the best variant within a family; robust to outliers
Q5-Q1 Sharpe	L1 + L2	Sharpe of (top-quintile - bottom-quintile) factor returns	L1: picks the best variant. L2: weighted 70% in family deployment composite
Rolling-window Sharpe	L2	Sharpe of the family's realized signal returns over its rolling backtest	Component of L2 robustness composite
t-statistic	L2	Statistical significance of the family's mean signal return	Component of L2 robustness composite
Deflated Sharpe Ratio (DSR)	L2	Sharpe adjusted for multiple-testing bias across all candidate families	Component of L2 robustness composite; also subtracted separately as an additional DSR penalty

Indicator Family Specifications

Brief operational descriptions of each family. Full parameter combinations and rolling-window behavior are governed by the selection methodology above.

- **Mean Reversion:** cross-sectional rank or z-score of recent return strength, with parameters L (lookback days, e.g. 20–120) and S (skip recent days, e.g. 10 or 20). Captures the “what goes up must come down” intuition.
- **SMA Cross / EMA Cross:** short-window MA vs long-window MA. Positive spread = uptrend; negative = downtrend. SMA is smoother; EMA reacts faster.
- **RSI:** momentum thermometer in [0, 100]. RSI > 70 = overbought; RSI < 30 = oversold; RSI = 50 = balanced.
- **MACD:** trend acceleration via Fast EMA – Slow EMA, with signal line and histogram. Histogram flips signal trend deceleration.
- **Bollinger Bands:** price position within a volatility channel (20-day MA $\pm$ 2 standard deviations). Touching the upper/lower band signals overextension; channel narrowing signals an upcoming volatility expansion.

- **Volume Spike:** today's volume vs rolling 20-day average. Spike threshold typically 3×. Signal of “something is happening” — must be combined with price direction.
- **Price Appreciation:** pure price momentum or price + volume confirmation.
- **Z-Score vs MA50:** standardized distance from the 50-day moving average, accounting for volatility.
- **Momentum:** trend persistence and continuation, parameterized by momentum window and return strength threshold.

Backtesting

The IOSG ETF Fund Strategy is validated across two independent benchmark categories. All series are backtested **Jan 2020 – Apr 2026 without survivorship bias, net of estimated fees:**

1. **Headline performance vs major competitor indexes** — head-to-head product comparison against Pure MCAP, Grayscale 5, Bitwise 10, Galaxy 25, and Bitcoin Buy & Hold.
2. **Universe variant robustness** — same Rt engine applied across 6 universe configurations (2 ranking methods × 3 exchange filters) to confirm the production variant is not cherry-picked.

Headline Performance vs Competitor Indexes

The production configuration is **Top 15 MCAP, Binance or Coinbase listed, Rt + 20% Cash**. To make the source of outperformance unambiguous, performance is shown in two charts:

1. Rt itself outperforms — before any cash overlay. Comparing the pure Rt-weighted portfolio (100% allocated, no cash) against five benchmarks under the same universe filter shows that Rt is already the highest-NAV strategy across the full backtest window. The edge comes from the variance-adjusted MinVar construction with liquidity and technical signal layers, **not from holding cash**.

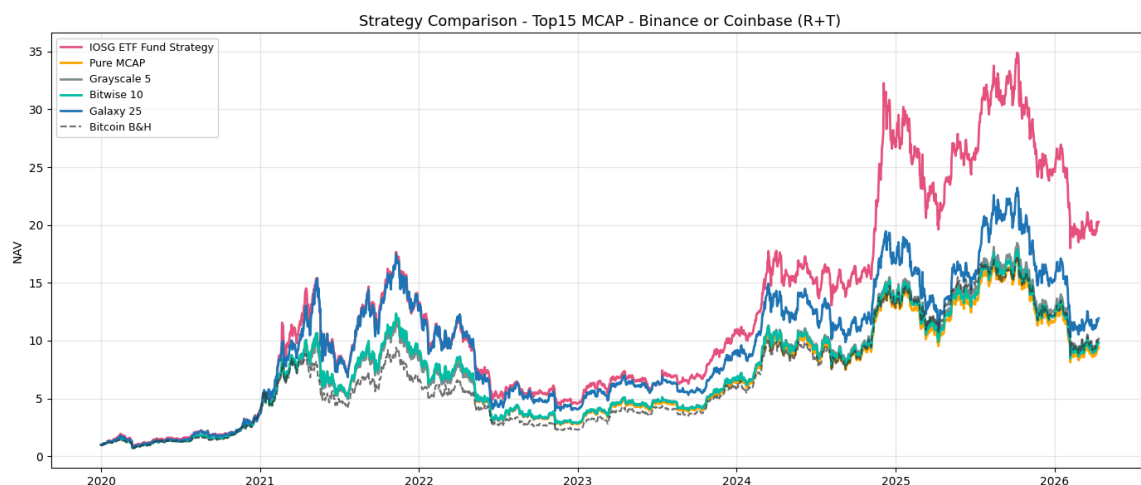


Figure 1: Strategy NAV Comparison — Top 15 MCAP, Binance or Coinbase, Rt only (no cash), Jan 2020 – Apr 2026

2. The 20% cash overlay adds incremental drawdown protection on top. Applying the production 20% cash buffer to Rt further improves Sharpe, reduces max drawdown, and lowers volatility (as quantified in *Cash Overlay* above). The chart below tracks NAV growth (rebased to 1.0 at Jan 2020) against the same five benchmarks, this time with all strategies including the 20% cash overlay for an apples-to-apples comparison.

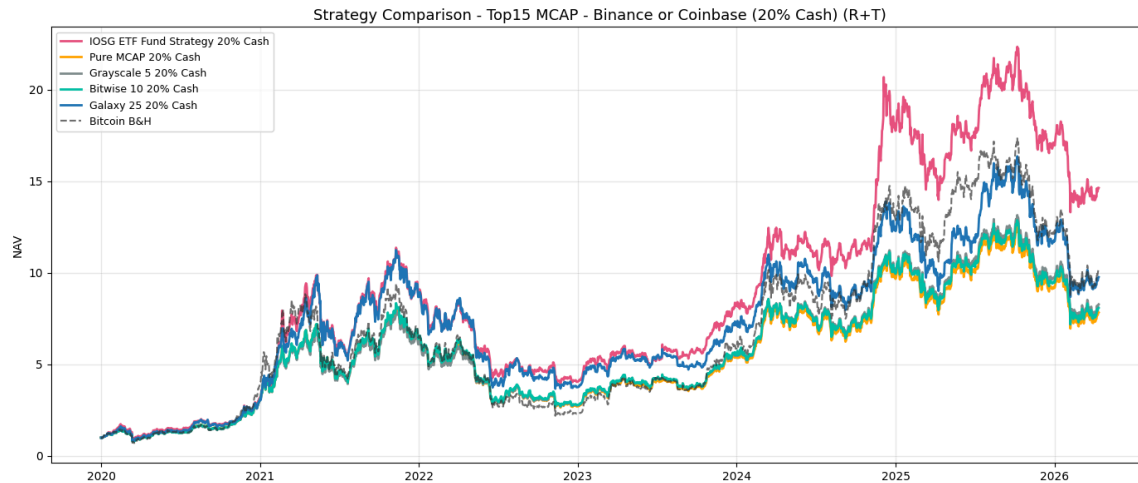


Figure 2: Strategy NAV Comparison — Top 15 MCAP, Binance or Coinbase, Rt + 20% Cash, Jan 2020 – Apr 2026

Returns & Efficiency

Strategy			Total Return	Annualized Return	Sharpe	Sortino
IOSG	ETF	Fund				
Strategy ★			1,366%	53.3%	1.02	1.21
Pure	MCAP	+ 20%				
Cash			685%	38.8%	0.76	0.93
Grayscale	5	+ 20%				
Cash			729%	40.0%	0.79	0.98
Bitwise	10	+ 20% Cash	710%	39.5%	0.77	0.95
Galaxy	25	+ 20% Cash	878%	43.8%	0.80	0.97
Bitcoin	Buy & Hold		915%	44.6%	0.73	0.92

Risk Metrics — Drawdown & Tail Loss

Strategy	Max Drawdown	Annualized Volatility	VaR (95%)	CVaR (95%)
IOSG ETF Fund Strategy ★	−65.3%	52.2%	−4.07%	−6.63%
Pure MCAP + 20% Cash	−68.0%	51.2%	−4.13%	−6.44%
Grayscale 5 + 20% Cash	−66.6%	50.6%	−4.05%	−6.31%
Bitwise 10 + 20% Cash	−67.6%	51.0%	−4.09%	−6.40%
Galaxy 25 + 20% Cash	−67.4%	55.0%	−4.50%	−6.97%
Bitcoin Buy & Hold	−76.7%	61.1%	−4.80%	−7.52%

The IOSG ETF Fund Strategy produces **+40% better Sharpe than Bitcoin Buy & Hold** (1.02 vs 0.73) and the lowest Max Drawdown across all six strategies tested (−65.3% vs Bitcoin’s −76.7%) — combining the highest risk-adjusted return with the strongest downside protection.

Return Profile

Across the **76-month backtest period**, the IOSG ETF Fund Strategy posted positive returns in **58% of months**, with a best month of **+57.0% (Feb 2021)** and a worst month of **−25.9% (Jun 2022)**. Bull-cycle years (2020 / 2021 / 2024) delivered triple-digit annual returns; the 2022 bear-cycle drawdown of −54.6% was contained relative to Bitcoin’s −64.2% the same year.

A full month-by-month return table for 2020 – 2026 is provided in *Appendix: Monthly Return Detail*.

Annual Returns vs Bitcoin

Annual return decomposition isolates *when* Rt generates excess return — and *when* it does not. Excess return is concentrated in (i) the 2021 alt-driven bull phase and (ii) the 2022 bear cycle

(smaller drawdown). In Bitcoin-dominated up-moves (2020, 2023, 2024) the strategy participates with the broader market but under-performs spot Bitcoin — the trade-off for diversification and lower volatility.

Year	IOSG ETF	BTC Buy & Hold	Excess
2020	+221.0%	+301.98%	−81.1 pp
2021	+178.6%	+59.79%	+118.7 pp
2022	−54.6%	−64.21%	+9.6 pp
2023	+103.2%	+155.61%	−52.6 pp
2024	+115.7%	+121.31%	−5.4 pp
2025	−3.7%	−6.33%	+2.6 pp
2026 YTD*	−14.4%	−19.36%	+5.0 pp

* 2026 data through April 12, 2026. BTC year-end closes sourced from Binance BTCUSDT daily series; 2026 YTD computed from Dec 2025 close.

Performance by Market Regime

Aggregating monthly returns by cycle phase clarifies the strategy’s behavior. The Rt + Cash construction outperforms Bitcoin in three of five regimes — including both bear-cycle drawdowns — while underperforming during the 2023 – 2025 BTC-dominated period. The underperformance narrows sharply from the Recovery phase (−65.4 pp) to Bull 2 (−3.4 pp) as broader alt participation recovered.

Regime	Months	IOSG Annualized	BTC Annualized	Excess (pp)
Bull 1 — Jan 2020 to Nov 2021	23	+237.6%	+194.6%	+43.0
Bear — Dec 2021 to Dec 2022	13	−57.7%	−68.1%	+10.4
Recovery — Jan 2023 to Oct 2023	10	+77.6%	+143.0%	−65.4
Bull 2 — Nov 2023 to Sep 2025	23	+82.9%	+86.3%	−3.4
Drawdown — Oct 2025 to Apr 2026	7	−45.4%	−49.3%	+3.9

The pattern is consistent with the strategy’s design objective: **Sharpe maximization and drawdown containment, not Bitcoin out-performance in BTC-dominated bull markets.** Excess return concentrates in altcoin-driven rallies and bear-market protection; the strategy’s structural under-weight to single-asset Bitcoin exposure is the cost of its diversification benefit.

Universe Variant Robustness — 6 Tested Configurations

The same Rt + 20% Cash engine is applied across **2 ranking methods × 3 exchange filters**. The production configuration (Top 15 MCAP, Binance or Coinbase listed) is selected because it produces the highest Sharpe Ratio across all six tested variants — confirming the universe rule is not cherry-picked.

Label	Universe	Exchange Filter	Sharpe
Top 15 MCAP — No Filter + 20% Cash	Top 15 by market cap	None	1.01
Top 15 MCAP — Binance or Coinbase + 20% Cash ★	Top 15 by market cap	Binance or Coinbase	1.02
Top 15 MCAP — Coinbase Only + 20% Cash	Top 15 by market cap	Coinbase listed only	1.00
Top 15 Liq (Top 100) — No Filter + 20% Cash	Top 15 by liquidity in Top 100 market cap	None	0.99
Top 15 Liq (Top 100) — Bn or Cb + 20% Cash	Top 15 by liquidity in Top 100 market cap	Binance or Coinbase	0.99
Top 15 Liq (Top 100) — Cb Only + 20% Cash	Top 15 by liquidity in Top 100 market cap	Coinbase listed only	0.93

★ Production configuration — selected based on Sharpe Ratio performance across all 6 tested variants.

Capacity Analysis

Strategy capacity is expressed relative to Bitcoin (BTC = 100 baseline) rather than absolute dollar figures. Absolute capacity depends on prevailing market depth at any point in time; the relative framing is more meaningful as a cross-strategy comparison.

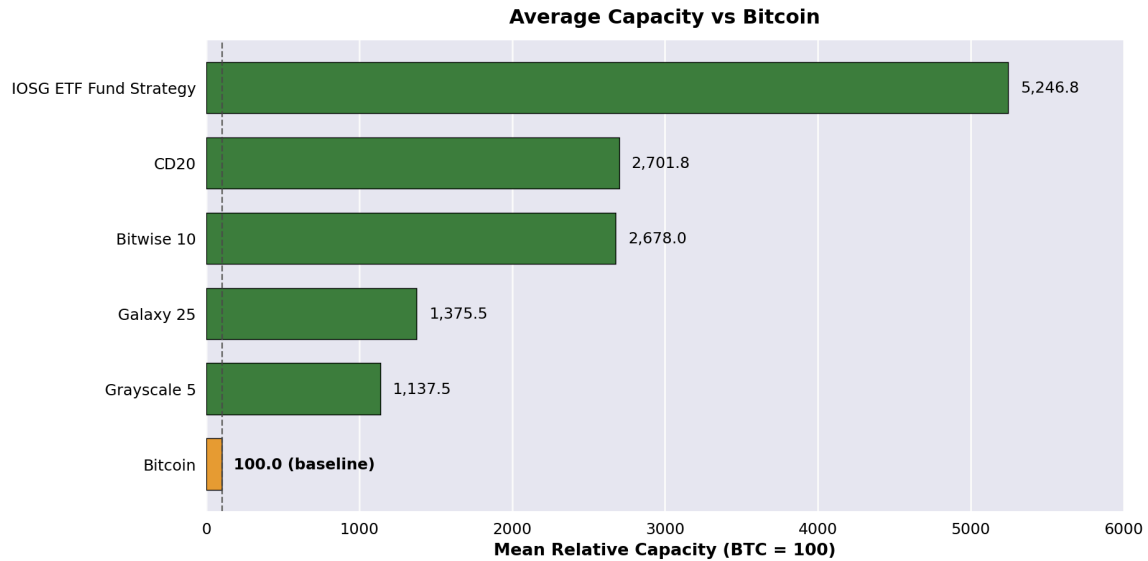


Figure 3: Average Capacity vs Bitcoin — IOSG ETF Strategy vs Major Crypto Indexes (BTC = 100 baseline)

Key Insights

- The IOSG ETF Fund Strategy shows approximately **4.6× higher capacity than Grayscale 5** (5,246.8 vs 1,137.5 relative to BTC).
- **CD20 and Bitwise 10 land at nearly identical capacity** (2,701.8 vs 2,678.0 relative to BTC).

The full computation framework, parameters, and analysis window are detailed in *Appendix: Capacity Calculation Methodology*.

Appendix

Portfolio Optimization

Objective Function

Portfolio weights are derived monthly by solving a constrained optimization problem:

$$\min_w w^\top \Sigma_{\text{adj}} w - k \cdot \sum_i \log w_i$$

Where:

- w is the **weight vector** — one weight per asset, summing to 100%
- Σ_{adj} is the **adjusted covariance matrix** — historical covariance with diagonal variances inflated/deflated by the liquidity penalty (Stage 3) and the technical signal adjustment (Stage 4)
- $\sum_i \log w_i$ is the **entropy barrier** (log-barrier) — diverges to $-\infty$ as any weight approaches zero, so subtracting it penalizes concentration and pushes the optimizer away from collapsed weights
- k is the **entropy coefficient** — a small positive scalar ($\sim 2e-7$), calibrated dynamically each rebalance to nudge diversification without overriding variance minimization

The **variance term** $w^\top \Sigma_{\text{adj}} w$ measures total portfolio risk and is what the optimizer is primarily minimizing. The **barrier term** $k \cdot \sum_i \log w_i$ is subtracted from the objective: it grows steeply whenever any single weight approaches zero, so subtracting it adds a penalty that prevents the optimizer from collapsing weight onto a few low-volatility names. The optimization is subject to the per-asset bounds, group floors and caps, and full-investment equality defined in *ETF Constraint Structure*.

Optimizer Selection

The solver is **Sequential Least Squares Programming (SLSQP)**, chosen because:

- It is the industry standard for constrained portfolio optimization.
- It handles equality and inequality constraints efficiently in a single call.
- It converges in 50–100 iterations for 15 assets.

At each iteration, SLSQP linearizes the constraint set and solves a local quadratic subproblem for the search direction. A line search determines the step size. The procedure repeats until the Karush–Kuhn–Tucker (KKT) optimality conditions are satisfied within a specified tolerance.

All constraints — per-asset bounds, group floors, group caps, and the full-investment equality — are passed directly to the solver. There are **no sequential post-hoc projections** in the ETF; the solver’s output is the final weight vector.

Optimization Pipeline

The pipeline proceeds through a sequence of stages. **Each stage modifies the covariance matrix or constraint set before the optimizer is called.**

Stage 1: Covariance Estimation

A 180-day trailing window of daily returns estimates the covariance matrix for the 15 selected assets. High positive covariance indicates assets move together; near-zero covariance indicates independent movement.

Why a 180-day lookback? Crypto markets are highly volatile and subject to regime shifts. Shorter horizons (30–90 days) produce unstable covariance estimates; 180 days reduces estimation error and captures more representative behavior.

Stage 2: L2 Ridge Regularization

A small **trace-normalized** ridge term is added to the covariance diagonal:

$$\Sigma_{\text{ridge}} = \Sigma_{\text{raw}} + \lambda \cdot \frac{\text{tr}(\Sigma_{\text{raw}})}{n} \cdot I, \quad \lambda = 0.001$$

The $\text{tr}(\Sigma)/n$ scaling makes the ridge term proportional to the matrix's average diagonal variance, keeping $\lambda = 0.001$ effective across both high- and low-volatility regimes. This stabilizes the estimate when the number of assets approaches the number of observations and prevents near-singular behavior during optimization.

Stage 3: Liquidity Penalty

The penalty inflates the diagonal variance of illiquid assets so the optimizer naturally reduces their weight. For each asset at the rebalance date, compute the rolling 30-day median trading volume and rank all 15 assets by volume percentile. Only the bottom 50% receive a penalty.

The variance multiplier is a linear function:

- Lowest percentile (rank = 0): multiplier = 3.0 (maximum inflation)
- 50th-percentile cutoff: multiplier = 1.0 (no penalty)
- Above the 50th percentile: multiplier = 1.0 (no penalty)

Only diagonal variances are adjusted. Off-diagonal covariances are preserved.

Stage 4: Technical Signal Adjustment

Technical signals modify the diagonal variance via a quantile-based multiplier system. At each rebalance, the framework recomputes technical scores for all 15 assets using the family-rotation mechanism, then converts scores to a rank-percentile map (0 = lowest, 1 = highest):

- **Top set** — assets at or above the 75th percentile, capped at 4 names. Multiplier < 1.0 (variance reduced; allocation increased).
- **Bottom set** — assets at or below the 25th percentile, capped at 4 names. Multiplier > 1.0 (variance inflated; allocation reduced).
- **Middle set** — multiplier = 1.0 (no change).

Volatility-Adaptive Calibration

The penalty intensity is dynamically scaled via a volatility-adaptive search that matches a target turnover between the risk-only and risk-plus-technical weight vectors:

- Compute current portfolio volatility from a market-cap-weighted reference portfolio over a trailing 6-month window.
- Map current volatility to its historical percentile within the stored volatility history. This produces a target impact (turnover) ranging from a floor of 1% to a ceiling of 20%, with a base of 10%.
- Search over scaling parameter $\alpha \in [0.0, 2.0]$ in steps of 0.02. For each candidate, the top-set penalty is $1.0 - \alpha \times 0.90$ and the bottom-set penalty is $1.0 + \alpha \times 1.00$.
- Build the penalty vector, apply it to the covariance diagonal, solve the full SLSQP optimization with all constraints, and measure realized turnover against the risk-only weights. Select the α whose turnover is closest to the volatility-derived target.

In high-volatility regimes, the historical percentile is high, producing a larger target impact and a larger α — technical signals exert more influence. In low-volatility regimes, the spread is dampened.

Stage 5: Entropy Calibration

An entropy barrier is added to the objective function to discourage extreme concentration. The entropy coefficient is calibrated dynamically at each rebalance via binary search:

- Solve the optimizer without entropy (pure minimum variance).
- Solve with a trial entropy coefficient.
- Measure the weight difference (half the sum of absolute changes).
- Adjust the coefficient until the difference matches a target turnover of 10%.
- Repeat for up to 50 iterations with 0.5% tolerance.

ETF Constraint Structure

Universe construction is defined in *Rt Universe* above (Binance or Coinbase listing, risk-flag exclusions, top 15 by previous-day market cap, with the optional crypto-adjacent stocks overlay).

The SLSQP solver receives all constraints in a single call:

Equality constraint: - All weights sum to 100% (fully invested)

Per-asset bounds (tiered by market-cap rank, mutually exclusive): - **Bitcoin (rank 1):** floor 1%, cap 50% - **Top 3 excluding Bitcoin (ranks 2–3):** floor 1%, cap 20% - **Mid tier (ranks 4–8):** floor 1%, cap 15% - **Bottom tier (ranks 9–15):** floor 1%, cap 10%

Group inequality constraints: - **Top 3 aggregate floor:** combined weight of Bitcoin and the next two largest assets (ranks 1–3) $\geq 50\%$. This is a *group* floor — it does not raise any individual asset's cap. - **Stock aggregate cap** (*only when overlay is active*): equity-linked assets $\leq 30\%$ combined

A fixed 20% post-optimization cash overlay is then applied to the SLSQP weights, as detailed in *Cash Overlay* above.

Why 1% floor and tiered caps (50% / 20% / 15% / 10%)? Prevents weight concentration while allowing flexibility for genuine high-conviction positions like Bitcoin. The 1% floor ensures every asset earns a non-trivial allocation rather than being squeezed to zero by the optimizer; the tiered

caps preserve broad exposure to mid- and bottom-tier names instead of letting Bitcoin and the top-3 absorb the entire portfolio.

Layered Covariance Adjustment

In the ETF, the covariance matrix passes through **two adjustment layers** before reaching the optimizer:

Raw Covariance

- Liquidity Penalty (Stage 3)
- Technical Penalty (Stage 4)
- Optimizer (L2 Ridge + Entropy applied inside the solver)

An asset that is liquid and technically strong presents the lowest effective variance and receives the highest allocation. An illiquid asset with poor momentum is pushed toward its 1% minimum bound.

Critical Parameters Summary

Parameter	Value	Effect
Min individual weight	1%	Forces broader diversification by preventing collapse to zero
Tiered max weights (rank 1 / 2–3 / 4–8 / 9–15)	50% / 20% / 15% / 10%	Per-asset caps tiered by market-cap rank (mutually exclusive)
Top 3 aggregate floor	$\geq 50\%$	Group constraint on combined weight of ranks 1–3
Stock aggregate cap	$\leq 30\%$	Active only when crypto-adjacent stocks overlay is on
Lookback period	180 days	Covariance estimation window
Liquidity volume window	30 days	Rolling median dollar volume
Liquidity penalty K	3	Variance multiplier at the lowest-volume token (rank 0); the multiplier is 1.0× (no penalty) above the p50 cutoff
L2 ridge λ	0.001	Trace-normalized perturbation to keep Σ invertible
Entropy coefficient k	2e-7	Default starting value; dynamically calibrated each rebalance

Capacity Calculation Methodology

Strategy capacity is computed using an **additive approach** for consistent cross-strategy comparison. For each token in the portfolio, the maximum dollar exposure consistent with absorbing only a fixed fraction of daily volume is divided by that token’s portfolio weight, then summed across all constituents:

$$\text{Token Capacity}_i = \frac{0.001 \cdot V_i^{30d}}{w_i}$$

$$\text{Total Portfolio Capacity} = \sum_i \text{Token Capacity}_i$$

Where V_i^{30d} is the 30-day median trading volume of token i and w_i is its portfolio weight at the rebalance date.

Key Parameters

Parameter	Value
Volume capture per token	0.1% of daily volume
Lookback window	30-day median volume
Analysis period	2020 – 2026 (75 rebalance points)

The 0.1% volume-capture assumption is the conservative baseline used throughout the cross-strategy comparison. Capacity scales linearly with this parameter, so the relative ranking across strategies is invariant to its choice within reasonable bounds.

Monthly Return Detail

Month-by-month returns for the IOSG ETF Fund Strategy (Top 15 MCAP, Binance or Coinbase, Rt + 20% Cash). Backtested, net of estimated fees. 2026 data through April 12, 2026.

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	YTD
2020	40.5%	0.7%	-25.1%	31.9%	5.6%	-3.5%	23.2%	9.4%	-8.9%	5.6%	38.0%	25.9%	+221.0%
2021	27.9%	57.0%	17.9%	13.8%	-23.4%	-4.8%	8.1%	27.2%	-9.4%	31.1%	0.0%	-13.2%	+178.6%
2022	-20.8%	-1.7%	21.0%	-16.3%	-14.8%	-25.9%	14.5%	-9.7%	2.6%	3.8%	-14.4%	-3.2%	-54.6%
2023	26.2%	2.1%	7.4%	2.3%	-2.6%	-6.5%	10.7%	-3.2%	2.9%	13.5%	9.1%	15.3%	+103.2%
2024	-1.2%	27.0%	18.8%	-8.6%	4.1%	-5.9%	5.9%	-5.5%	7.4%	0.2%	50.2%	0.0%	+115.7%
2025	7.5%	-16.2%	-4.1%	6.3%	6.7%	1.7%	15.2%	-1.3%	3.4%	-6.4%	-11.0%	-1.5%	-3.7%
2026*	-4.3%	-15.5%	1.7%	4.1%	—	—	—	—	—	—	—	—	-14.4%

* 2026 data through April 12, 2026. All returns are backtested simulations — not indicative of future results.